Lecture 5 - The Derivative

Garrett Credi

09/22/26

Homework Updates/Info/Yappery?

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3. If you're choosing homework problems in a way that's making you feel lost or making you feel unprepared for the quizzes please talk to me so we can discuss different strategies :)

Quiz Time!

Let’s get into nice orderly rows, spread out as much as possible, and slay this quiz :)

The Road So Far

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3. We developed *maaaany* different tools to compute limits.

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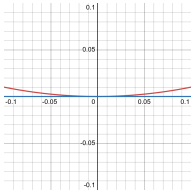


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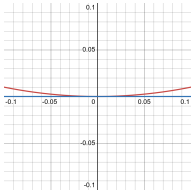


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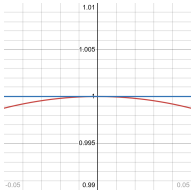


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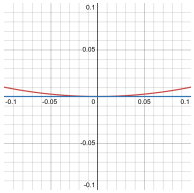


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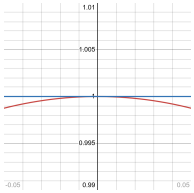


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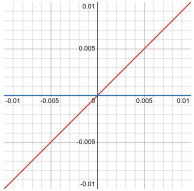


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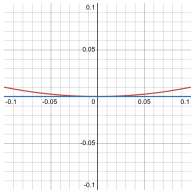


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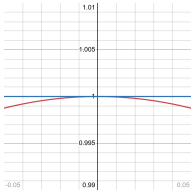


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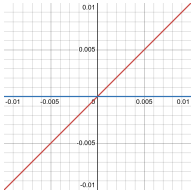


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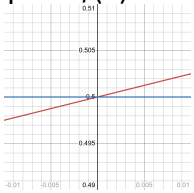


Figure: The graph of a logistic function near $x = 0$

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A linear function!

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3. The slope of the *tangent* line to the graph of $y = f(x)$ at $x = a$.

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Derivative of a function at a point

The derivative of a function $f(x)$ at a point $x = a$, denoted $f'(a)$ is the common value of the following four (equal) quantities:

1. $$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{4}$$
2. The slope of the best linear approximation to $f(x)$ at $x = a$.
3. The slope of the *tangent* line to the graph of $y = f(x)$ at $x = a$.
4. The instantaneous velocity of a particle with position described by $f(x)$ at time $x = a$.

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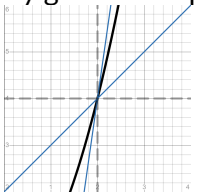


Figure: Linear approximations of $f(x) = x^2$ around $x = 2$ with slopes $m = 1, 7$

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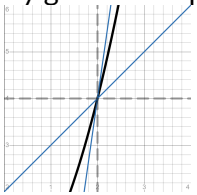


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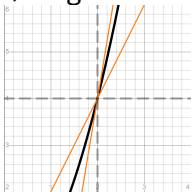


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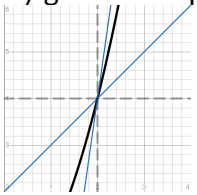


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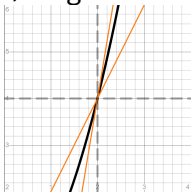


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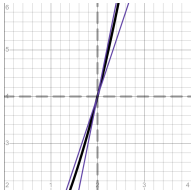


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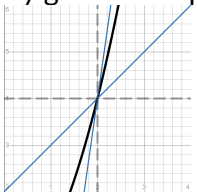


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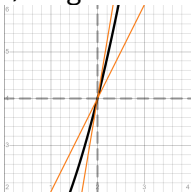


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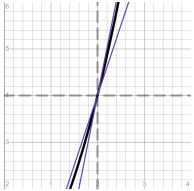


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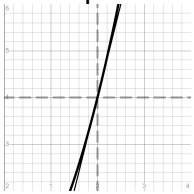


Figure: The best linear approximation of $f(x) = x^2$ around $x = 2$ with slope $m = f'(2) = 4$.

Computing the Derivative II

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$$= \lim_{h \to 0} \frac{(4 + h) - 4}{(\sqrt{4 + h} + \sqrt{4})h}$$

Computing the Derivative II

Let's try another one! Let's compute $f'(4)$ when $f(x) = \sqrt{x}$.

$$f'(4) = \lim_{x \to 4} \frac{f(x) - f(4)}{x - 4} \tag{14}$$

$$= \lim_{x \to 4} \frac{\sqrt{x} - \sqrt{4}}{x - 4} \tag{15}$$

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$$= \lim_{x \to 4} \frac{(\sqrt{x})^2 - (\sqrt{4})^2}{(x - 4)(\sqrt{x} + \sqrt{4})} \tag{17}$$

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$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0}$$

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$$\begin{aligned} f'(0) &= \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \to 0} \frac{|x|}{x} \end{aligned} \tag{25}$$

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Since $\lim_{x \to 0^+} \frac{f(x) - f(0)}{x - 0} \neq \lim_{x \to 0^-} \frac{f(x) - f(0)}{x - 0}$

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Since $\lim_{x \to 0^+} \frac{f(x) - f(0)}{x - 0} \neq \lim_{x \to 0^-} \frac{f(x) - f(0)}{x - 0}$ this means that $f'(0)$ *does not exist* and there is ***NO*** best linear approximation to $|x|$ around $x = 0$.

Differentiable Functions I

Differentiable Function

A function $f(x)$ is *differentiable* at $x = a$ if any of the following equivalent conditions hold:

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2. There exists a best linear approximation $\mathscr{L}(x) = f(a) + f'(a)(x - a)$ to $f(x)$ around $x = a$.
3. There is a unique tangent line to the graph of $y = f(x)$ at $x = a$.

Differentiable Functions II

The two classes of functions we've defined so far

Continuity and Differentiability

Differentiable Functions II

The two classes of functions we've defined so far (continuous & differentiable)

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For a function $f(x)$ and a point $x = a$ we have the following (equivalent) relationships:

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The following intuition may be useful:

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We can intuitively recognize continuity/differentiability of a function $f(x)$ at a point $x = a$ from its graph as:

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2. differentiability: if the graph of $f(x)$ "locally looks like" the graph of a linear function near $x = a$ i.e. the function is both continuous and does not make any sharp changes near $x = a$.

Terminological Potpourri I

There's a lot of terminology so here's some scattered parts that didn't fit nicely.

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If our function represents the relationship between

E.g. if $s(t)$ is the distance in miles that a car travels, where t is the time since starting measured in hours,

Terminological Potpourri II

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Terminological Potpourri II

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E.g. if $s(t)$ is the distance in miles that a car travels, where t is the time since starting measured in hours, $s'(t) = v(t)$ is the velocity and is measured in units of miles per hour.

Worksheet Time!

Let’s get into groups and work on the worksheet!

End of Class!

That’s all folks!